

Risk Assessment and Mitigation

CO3 G10

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Risk Management Process

At the beginning of the project, we identified risks and placed them in the categories: project, product, project and product, and business risks. A Google Sheets document was used to track these risks and assign them a likelihood and severity of either low, moderate, or high. Mitigation strategies for each risk were included. Throughout the project, risk has been considered and added to the register as identified. During week 5, improvements were made to the risk register format to include more details such as a risk ID and owner. Throughout the project, risks can be added to the register as identified.

Risk Register Format

The current register format is a table containing the following headings, ID, Type, Description, Likelihood, Severity, Mitigations, and Owner.

ID - A unique identifier used to refer to the risk throughout the project documents.

Type

The element of the project the risk is related to, we have used the following types:

- Project: risks to schedule and team resources.
- Product: risks to the quality of the finished product.
- Business: risks to the customer's business and ability to market the product.

Some risks can fall into multiple of these categories.

Description - A short description of the risk.

Likelihood - A rating of low, moderate, and high where low represents a risk that is very unlikely to happen and high means very likely. The colours green, orange, and red are used respectively to highlight more likely risks.

Severity - A rating of low, moderate, and high where low represents a risk that will have a minimal impact and high means high impact. The colours green, orange, and red are used respectively to highlight more severe risks.

Mitigations - a description of what can be done to mitigate the risk.

Owner - the person or group responsible for monitoring and mitigating against the risk.

ID	Type	Description	Likelihood	Severity	Mitigations	Owner
R1	Project	Team members not contributing enough	Moderate	Moderate	Keep in contact with everyone. Make sure tasks are assigned to everyone, and that they're being completed. Project tracking tools.	JT
R2	Project	Team members not being happy in their role.	Moderate	Low	Make sure everyone is communicating about their work, and raise any concerns with the team. Reassign if needed.	WB
R3	Project	Team members in charge of implementation using low spec laptops	Moderate	Moderate	Make 3 members of our group in charge of implementation (likelihood of all having low spec computers is low), if worst case happens - they can use the computer lab pc's	WB, JT
R4	Project	Programmer or Documentation member becomes unavailable	Moderate	Moderate	Ensure that there are at 3 people in charge of Programming/implementation and at least 4-5 in charge of documentation so that there is always an extra person to step in, in case of absences/unforeseen circumstances	CH
R5	Project	A situation where a sub group in our team can't do any work until the other sub group completes their work (due to one group's work depending on another)	High	Moderate	Ask the other sub group to help the sub group doing the work to speed up efficiency and productivity within the group (if no alternative work can be found for them)	CH
R6	Project	Some group members may only be available remotely (due to inability to commute weekly)	Low	Moderate	Make sure that they are regularly contacted online, assign them tasks and make it clear to them who is in their sub group, make sure that the tasks assigned to them take into account that they won't be attending any in person group meetings	JT
R7	Product	Bugs in our implementation	High	Moderate	Regular testing should be done, other implementation	DK, SM

					members should also test before making changes to make sure that everything functions well	
R8	Product	Randomly generated maze is impossible to be solved on time	Moderate	Moderate	Make it clear to the implementation team that safeguards should be put in place within the code to make sure that the maze that is generated is actually solvable. Regular testing whilst programming is essential to making sure these requirements are met.	OS
R9	Product	Game incompatibility with certain computer systems	Moderate	High	Contact the library owners/creators for support if unexpected incompatibility persists (make sure that the libraries are up to date in the meantime), Research alternative libraries that are widely used and hence compatible with multiple systems	OS, SM
R10	Product, Project	Bugs in 3rd party libraries	Low	High	Compile a list of alternative game engines to use in order to deal with unexpected setbacks with the current library. There are popular, reliable and widely used alternative libraries available	DK
R11	Product, Project	Language barriers between group members	Low	Moderate	Make sure that the student fully understands the most important parts of the project such as the user/system requirements, use a reliable language translator such as google translate whenever the student doesn't fully understand what is being said to make sure that the proper message is conveyed	JT
R12	Product, Project	Low attendance and zero cooperation from a significant number of group members	Moderate	High	Try to communicate with the missing people, contact the relevant lecturers to raise any issues	JT, WB
R13	Product,	Most of the team	Low	Moderate	Try to avoid	JT

	Project	members not getting along with each other e.g regular conflicts/arguments			conflicts/arguments within the group, show a willingness to compromise on what tasks each member wants to do, make the group leader intervene and have the final say if conflicts persist, get the lecturers involved if no resolution has happened,	
R14	Product, project	Data loss	Low	High	Ensure everything is uploaded onto google drive and GitHub.	WB
R15	Product, project	Establishing incorrect requirements	Low	High	Make sure the interview covers all aspects of the game in detail. Make the customer aware of any assumptions made.	CH
R16	Product, Project	Client's schedule for interviews (after the first interview) is not flexible	Low	Moderate	Make sure that the first interview is in depth and extensive, covering all possible requirements and questions that can be asked	WB
R17	Product, Project	Once a feature is implemented, it might be clear that it may not be as effective in solving requirements as initially thought	Moderate	Moderate	Make sure there are multiple versions, so we can revert back to a previous version if the feature isn't as effective, draw up a list of contingency features that can be implemented instead	OS, DK
R18	Business	Competition from other groups (during the presentation)	High	Moderate	Make sure that the game is simple, and that it follows the requirements stated by the client, make sure that the game has features that will no doubt make it stand out in comparison to other groups work	WB, JT
R19	Business	The risk that other groups would finish their projects quicker than us and use that extra time to build more superior features due to us using outdated/less superior online software/tools	Moderate	Moderate	Make sure that the tools/software that are used in designing, implementing and documenting the game are good, up to date and reliable enough to use to make sure that we don't spend unnecessary time trying to do a task that can be done quicker using an alternative tool	WB